

NAME SOLUTIONS

ECE 4644 / 5610 – Satellite Communications

Mid-term

Wednesday, March 18, 4:00 – 5:15 PM

Instructions

This is a closed-book, in-class exam. A sheet of equations is provided. You are allowed only a writing implement and a calculator.

The exam must be turned in promptly at the end of the designated exam period or you will lose points.

First, write your name at the top of this page.

For numerical problems highlight your final answer by underlining or circling. *Important: Re-read the question after you have answered it to be sure that you have answered the question asked and not missed something.* Remember: it is important to show your working with a reasonable amount of explanation to gain full credit. Make sure everything is legible: use ink or a dark pencil. There is a generous amount of space for working out your answer to each problem but if you need more, use a blank side of an exam sheet.

If you think something is missing from a problem specification, assume a suitable value, state your assumptions clearly, and proceed with the problem. If you conclude that something will not work, say so, make suitable assumptions, and proceed with the problem.

All the rules of the Virginia Tech Graduate Honor System apply strictly to this exam. Any violation will result in confiscation of the exam and a grade of zero.

Grading

Your score will be the sum of the points you earn for each section. This will be your grade out of 60, to a maximum of 60.

Section 1 26 /26

Section 2 30 /30

Section 3 10 /10

Total 60 /60

Section 1: (26 points available, 2 point per answer)

Instructions: Answer the following questions by clearly circling the correct answer. You may support your answer with a brief explanation. If no answer seems appropriate, make a note to that effect. Partial points may be awarded.

1. The following is true of a satellite in geostationary orbit:
 - a. the period of its orbit is 24 hr
 - ☒ b. the period of its orbit is one sidereal day
 - c. the inclination of its orbit is 90°
 - d. the ascending node is in the direction of the First Point of Aries
2. The proper ordering of orbits in terms of *lowest-to-highest* altitude is:
 - a. GEO > LEO > MEO
 - b. MEO > GEO > LEO
 - c. LEO > GEO > MEO
 - ☒ d. LEO > MEO > GEO
3. The following uplink / downlink frequency combination is a candidate for a C-band system:
 - a. 50 / 40 GHz
 - b. 30 / 20 GHz
 - c. 14 / 11 GHz
 - ☒ d. 6 / 4 GHz

C band: 4 - 8 GHz
4. Radio signal propagating in free space with a frequency of 15 GHz has a wavelength of
 - a. 1 cm
 - ☒ b. 2 cm
 - c. 3 cm
 - d. 6 cm

$f [\text{GHz}] \lambda [\text{cm}] = 30$
5. If the semi-major axis of a satellite orbit is increased by a factor of 4, the orbital period will increase by a factor of
 - a. 16
 - ☒ b. 8
 - c. 4
 - d. 2

$T_p \propto a^{3/2}$
 $\Rightarrow a \uparrow 4x \text{ then } T_p \uparrow 8x$
6. If a satellite orbit has an apogee distance that is four the perigee distance, then for velocity
 - a. $v_{\text{apog}} = 4v_{\text{per}}$
 - b. $v_{\text{apog}} = 2v_{\text{per}}$
 - c. $v_{\text{apog}} = 0.5v_{\text{per}}$
 - ☒ d. $v_{\text{apog}} = 0.25v_{\text{per}}$

Kepler's 2nd Law

7. For a satellite in LEO orbit, the primary source of orbital perturbation is
- ☒ a. atmospheric drag
 - b. non-spherical shape of Earth
 - c. gravity of the Moon and Sun
 - d. eclipse effects
8. In a communications receiver the input signal is at 14 GHz and the IF section requires a frequency of 2 GHz. For the case of *low-side* injection, a suitable LO frequency is
- a. 10 GHz
 - ☒ b. 12 GHz
 - c. 14 GHz
 - d. 16 GHz

$$14 \otimes 12 = \frac{26}{2} \text{ GHz}$$

9. For a standard receiver equipped with a high-gain LNA, the noise temperatures of these two devices will likely matter the most to the overall system noise temperature:
- ☒ a. antenna and LNA
 - b. LNA and mixer
 - c. mixer and IF amplifier
 - d. antenna and IF amplifier

$$T_s \approx T_{\text{ant}} + T_{\text{LNA}}$$

10. For a satellite in a LEO orbit with an inclination of 0° eclipses will occur
- ☒ a. every orbit
 - b. around the solstices
 - c. around the equinoxes
 - d. never

11. If the diameter of a circular satellite antenna is doubled, then its gain increases by
- a. 3 dB
 - b. 4 dB
 - ☒ c. 6 dB
 - d. 10 dB

$$G \propto A_p \propto D^2$$

$$\Rightarrow D \uparrow 2x \text{ then } G \uparrow 4x \text{ or } 6 \text{ dB}$$

12. At the moment of *southern* summer solstice, the R.A. and declination of the Sun are, respectively,
- a. 0 deg, 0 deg
 - b. 90 deg, $+23\frac{1}{2}$ deg
 - c. 180 deg, 0 deg
 - ☒ d. 270 deg, $-23\frac{1}{2}$ deg

The sun is at its greatest distance south of the celestial equator

13. If a satellite system has C/N values of 30 dB on both the uplink and the downlink, the overall C/N is
- a. 15 dB
 - ☒ b. 27 dB
 - c. 30 dB
 - d. none of the above

$$1/(C/N)_o = 1/(C/N)_{up} + 1/(C/N)_{dn}$$

$$30 \text{ dB} = 1000x$$

$$\Rightarrow (C/N)_o = 500 = 27 \text{ dB}$$

Section 2: (30 points available, 5 points per answer)

Problem #1 Project Kuiper satellites are being deployed at an altitude of 450 km.

- a) Find the length of an arc along Earth's surface passing through the subsatellite point on which an Earth station could communicate with the satellite, assuming this is possible to the horizon
- b) Project Kuiper operates near 30 GHz. Briefly discuss limits on elevation angle

a)



$h = 450 \text{ km}$

$$\cos \theta = \frac{R_E}{(R_E + h)} = \frac{6378.137}{(6378.137 + 450)}$$
$$\Rightarrow \theta = 20.9^\circ = 0.365 \text{ rad}$$

The length of the arc is $2 = 2R_E \theta$

$$= 2(6378.137)(0.365)$$
$$= \underline{4656 \text{ km}}$$

- b) The preceding analysis assumes operation down to 0° elevation angle.

30 GHz is Ka band, which generally requires $el \geq 20^\circ$. Given this consideration and blockage by terrain and buildings, the arc of visibility will be considerably smaller.

Problem #2 GPS satellites fly in circular orbits at a height of 20,200 km:

- a) find the period of the orbit in seconds and hr:mn:sc format
- b) determine the velocity of the satellite in its orbit

a) Circular orbit so $a = R_E + h = 26,578.137 \text{ km}$

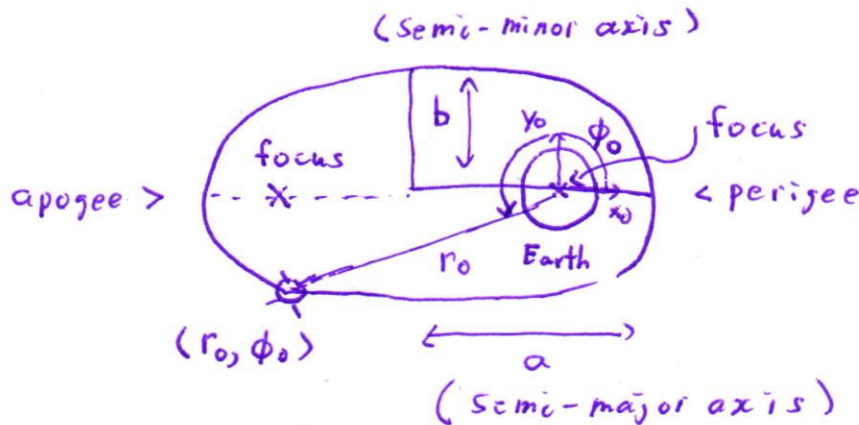
$$\text{Then } T_p = (2\pi a^{3/2}) / \mu^{1/2} = (2\pi [26,578.137]^{3/2}) / [3.9860044 \cdot 10^5]^{1/2}$$
$$= \underline{43121.89 \text{ sec}}$$
$$= \underline{11^{\text{hr}} 58^{\text{mn}} 41.9^{\text{sc}}}$$

b) $v = \frac{2\pi a}{T_p}$

$$= \frac{2\pi (26,578.137)}{43121.89} = \underline{3.87 \text{ km/sec}}$$

Problem #3 Sketch an ellipse corresponding to a satellite orbit with significant eccentricity and indicate the following:

- the positions of the foci of the ellipse (roughly), Earth, and the semi-major and semi-minor axes
- the positions of perigee and apogee
- the satellite at a point in its orbit that is more than half a period after a time of perigee and indicate how this position is mapped by the orbital plane coordinates (r_o, ϕ_o)



Problem #4 A MEO satellite communications system transmits from an earth station with a power of 500 W and an antenna gain of 30 dB over a distance of 4000 km.

- Find the power density incident on the satellite antenna assuming perfect alignment
- Determine the maximum power that could be available at the output of the satellite receiving antenna if it has an effective area of 3 m²
- Name two loss factors that will likely result in the power being lower

i) $P_t = 500 \text{ W}$
 $G_t = 30 \text{ dB}$

$R = 4000 \text{ km}$

$$F = \frac{P_t G_t}{4\pi R^2} = \frac{(500)(1000)}{4\pi (4104)^2}$$

$$= 2.487 \cdot 10^{-9} \text{ W/m}^2$$

ii) $P_{max} = P_r = F A_e = (2.487 \cdot 10^{-9})(3) = 7.460 \cdot 10^{-9} \text{ W}$

iii) Loss factors:

- misalignment of the antennas
- atmospheric absorption including weather factors

Problem #5 Give the names of the three orbital elements (i , Ω , ω) and briefly explain each. State the range of possible values for each element as encountered with satellite orbits.

Orbital elements:

- i : inclination angle describes the tilt of the orbital plane wrt the equatorial plane. $i = 0^\circ - 90^\circ$ describes a prograde orbit while $90^\circ - 180^\circ$ describes a retrograde orbit
- Ω : right ascension of the ascending node (RAAN) gives the angle in the equatorial plane between the dirⁿ to the First Point of Aries (FPA) and the dirⁿ to the point where the satellite rises (going N) through the equatorial plane. Range: $0^\circ - 360^\circ$
- ω : argument of perigee gives the angle measured in the orbital plane between the dirⁿ to the FPA and the dirⁿ to perigee. Range: $0^\circ - 360^\circ$

Problem #6 Explain the meaning of AOCS and briefly describe the functions and workings of the AOCS subsystem of a communications satellite.

AOCS: Attitude and Orbit Control Subsystem

This satellite subsystem performs two functions: (i) attitude control maintains the orientation of the satellite so that it can keep its antennas and solar panels properly directed. Most satellites are three-axis stabilized with reaction wheels and use thrusters or ion jets to make corrections, (ii) orbit control maintains the satellite in the desired orbit. A satellite is perturbed in its orbit by forces, and maneuvers must be performed with jets or thrusters to counteract them. For a GEO satellite the principal perturbations are due to the gravitational pulls of the sun and moon (N-S effect on i) ~~and~~ and a non-uniform Earth's gravity (E-W effect on longitude) while for a LEO satellite the main perturbation is due to atmospheric drag.

Section 3: Design problem (10 points available)

We will design a receiving earth station to achieve a C/N of 20 dB on the downlink of a Ku-band LEO satellite communications system. Take the satellite to be passing directly overhead at a height of 1000 km. The downlink frequency is 15 GHz and the saturated power of the transponder is 100 W. Backoff of 3 dB is applied. The satellite transmitting antenna has a gain of 20 dB. The receiving antenna at the earth station is circular with an aperture efficiency of 60%. The system noise temperature of the receiver is 150 K and the noise bandwidth is 36 MHz. Allow 2.0 dB for antenna misalignment losses and 2.0 dB for clear air attenuation and miscellaneous losses.

- Find the noise power in the receiver, N , and then the signal power in the receiver, P_r , that is needed to make spec, both in dBW
- Calculate the path loss, L_p , in dB
- Arranging in the form of a link budget and including loss terms, all in dB units, determine the gain of the earth station receiving antenna that is needed to achieve the desired C/N
- Determine the diameter of the earth station receiving antenna that is required to make spec

$P_{sat} = 100 \text{ W}$
 $B_{ck} = 3 \text{ dB}$
 $G_t = 20 \text{ dB}$
 $h = 1000 \text{ km}$
 $f_{dn} = 15 \text{ GHz}$
 $\lambda = 2 \text{ cm}$

$a) N = k T_s B_n$
 $= (1.38 \cdot 10^{-23}) (150) (36 \cdot 10^6)$
 $= -131.3 \text{ dBW}$
 $C/N = 20 \text{ dB} \Rightarrow$
 $P_r = N + (C/N)$
 $= -111.3 \text{ dBW}$

$b) L_p = 10 \log \left(\frac{4\pi R}{\lambda} \right)^2$
 $= 10 \log \left(\frac{4\pi \cdot 10^6}{0.02} \right)^2 = 176.0 \text{ dB}$

$c) \text{ Link budget:}$

P_t	20 dBW
Backoff	- 3 dB
G_t	20 dB
G_r	G_r dB
L_p	-176 dB
L_{tot}	- 4 dB

$P_r = -111.3 = G_r - 143.0$

$d) G_r = -111.3 + 143.0$
 $= 31.7 \text{ dB}$

$d) G_r = \eta_A \frac{4\pi A_p}{\lambda^2}$
 $\Rightarrow A_p = \frac{G_r \lambda^2}{\eta_A 4\pi} = \frac{(1479)(0.02)^2}{(0.6) 4\pi}$
 $= 0.0785 \text{ m}^2 = \frac{1}{4} \pi D^2$
 $\Rightarrow D = \sqrt{\frac{(0.0785) 4}{\pi}} = 0.316 \text{ m}$

$L_{misalign} = 2.0 \text{ dB}$
 $L_{misc} = 2.0 \text{ dB}$
 $L_{tot} = 2.0 + 2.0 = 4.0 \text{ dB}$

$\eta_A = 60\%$
 $T_s = 150 \text{ K}$
 $N = 36 \text{ MHz}$
 $= 36 \cdot 10^6$